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Hardening the Ecosafety Stack: A Grammar-First Approach to Lowering Residual Risk Through Executable Contracts and Empirical Validation

The objective of this research is to develop a hardened Application Language Network (ALN) grammar framework that systematically lowers the risk-of-harm from corridor miscalibration and sensor errors by prioritizing canonical schemas and data-quality plane specifications. The central thesis is that the highest-leverage approach to risk reduction is not through incremental runtime improvements but through a foundational "grammar-first" strategy. This involves creating production-ready ALN artifacts, compile-time enforcement mechanisms, and cryptographic governance rules that make unsafe states non-representable [2]. The framework targets Phoenix qpudatashards as the anchor family due to their existing integration with KER and ecosafety corridors, while also providing a scalable model for Cyboquatic, FOG, and other ecological domains. This report details the design of core ALN particles like `DataQualityPlane2026v1.aln` and `EcoNetSchemaShard2026v2.aln`, outlines the implementation of schema-driven Rust macros, defines cryptographic provenance contracts, and presents a rigorous plan for empirical calibration and automated verification. The ultimate goal is to transform abstract safety principles into concrete, executable, and auditable contracts that form the backbone of a robust ecosafety stack.

Formalizing the Data Quality Plane with Executable Invariants

The cornerstone of the proposed grammar-first hardening strategy is the formalization of the data quality plane as a first-class, normalized risk coordinate within the system's grammar. This concept elevates `rcalib` (ingest/schema quality) and `rsigma` (sensor uncertainty) from being mere diagnostic fields to becoming integral components of the risk lattice, governed by the same strict rules as physical risks like PFAS or EMF [1]. The

primary artifact for this formalization is `DataQualityPlane2026v1.aln`, an ALN particle that binds these data-quality signals to the standard ecosafety corridor grammar, Lyapunov channels, and shard-level governance [3]. This section details the structure of this ALN particle, the critical invariants that translate high-level governance principles into machine-checkable contracts, and the process of calibrating the underlying mathematical kernels using real-world data. By making the behavior of `rcalib` and `rsigma` explicit, versioned, and enforced at every stage of the pipeline, this approach directly confronts the root causes of residual risk, such as misconfigured paths and hash misuse, by rendering them non-representable states in the system [11].

The `DataQualityPlane2026v1.aln` file serves as the grammar shell that pins `rsigma` and `rcalib` into a shared two-dimensional risk plane [4]. Its columns define the configuration of this plane for a given shard family, lane, and set of corridor definitions. The `plane_id` provides a unique identifier for the specific plane configuration, such as `DQ-PHX-2026-01` for the Phoenix data quality plane in 2026. The `rsigma` and `rcalib` columns themselves are declared as `RiskCoord01`, ensuring they are normalized floating-point values between 0.0 and 1.0, representing the degree of sensor uncertainty and ingest quality, respectively [7]. Critically, the plane is bound to the rest of the ecosafety stack through several foreign key and role columns. The `lyap_channel` is fixed to `dataquality`, dedicating a specific channel for these data-quality signals in the Lyapunov residual calculation [8]. The `corridor_table_id` links this plane to the master corridor table, typically `ecosafety.corridors.v2`. Within that table, the `rsigma_corridor_row` and `rcalib_corridor_row` columns point to the specific rows that define the safe, gold, and hard bands for each of these new risk coordinates, enforcing consistency with the established corridor grammar [9]. Furthermore, the plane specifies the contribution of these coordinates to the overall safety posture through `lyap_weight_rsigma` and `lyap_weight_rcalib`, which are non-negative weights applied to the quadratic terms in the Lyapunov residual, `V_t` [10]. Finally, the `lane` column dictates the governing rules for this plane, and the `admissible_for_blockchain` boolean acts as a gatekeeper, determining whether this specific plane configuration can be cryptographically anchored on the ALN-Blockchain [5].

The true power of `DataQualityPlane2026v1.aln` lies not just in its definition but in the set of ALN invariants that govern its contents. These invariants are the mechanism for turning prose-based governance policies into executable, compile-time checks, thereby making unsafe configurations impossible to represent. They ensure that any row in this ALN particle adheres to a strict set of mathematical and logical constraints. For instance, invariant `DQ1.rsigma_rcalib_range` enforces that both `rsigma` and `rcalib` must

be within the [0, 1] interval, preventing nonsensical values from propagating through the system. Invariant `DQ2.dataquality_channel` mandates that all data-quality planes must use the dedicated `dataquality` Lyapunov channel, preventing accidental mixing with physical risk channels. Invariant `DQ3.corridor_rows_exist` creates a crucial link by verifying that the referenced `rsigma_corridor_row` and `rcalib_corridor_row` actually exist within the specified corridor table and are correctly typed for the `dataquality` channel [6]. To prevent overly permissive rules, invariant `DQ4.standard_bands` ensures that the bands for `rsigma` and `rcalib` follow the established deep-safe, gold, and hard geometry, meaning `0.0 <= safemin < safemax <= goldmin < goldmax <= hardmin < hardmax <= 1.0` [7] [8].

More sophisticated invariants encode deeper safety properties. Invariant `DQ5.nonnegative_lyap_weights` is fundamental, requiring that both `lyap_weight_rsigma` and `lyap_weight_rcalib` be greater than or equal to zero. This mathematically enforces the principle that increased data uncertainty can only down-credit knowledge (`K`) and eco-impact (`E`), never inflate them; it guarantees that `V_t` is a monotonically non-decreasing function of `rcalib` and `rsigma` [10]. Invariant `DQ6.prod_hard_bands_strict` imposes stricter requirements for the PRODUCTION lane, mandating that the `hardmax` band for both `rsigma` and `rcalib` be no higher than 0.3. This ensures that data-quality risk remains a strong penalizer even when physical risks appear benign, preventing deployments based on low-quality data [9]. The most advanced invariant, `DQ7.blockchain_admissible_requires_monotone_ker`, directly ties the on-chain admission process to formal proof obligations. It stipulates that for a plane configuration to be marked as `admissible_for_blockchain = true`, there must exist a corresponding proof in a `ker_proofs.v1` ALN table that certifies the monotonicity of the entire KER system with respect to `rcalib` and `rsigma`. This turns a theoretical requirement for safety into a practical, cryptographic gatekeeper for the ledger [3]. Finally, invariant `DQ8.unique_plane_per_context` prevents governance ambiguity by ensuring that only one admissible data-quality plane exists for any given `(lane, region, shard_family)` combination, avoiding conflicting rules [5].

Once the grammar and invariants are frozen, the next step is to populate the plane with calibrated, empirically-grounded mathematics. The internal kernels for `rcalib` and `rsigma` are already defined in the corpus, and this phase focuses on fitting their parameters to real-world data from the Phoenix basin. For `rcalib`, the process begins with the ingest error kernel, `I`, which is a weighted sum of diagnostic counts: $I_{\text{rcalib}} = w_{\text{miss}} N_{\text{miss}} + w_{\text{rfc}} N_{\text{rfc}} + w_{\text{type}} N_{\text{type}} + w_{\text{varid}} N_{\text{varid}}$ [1]. The research action here is to use historical Phoenix ingest logs to perform a regression analysis, correlating

different combinations of fault counts (`N`) with observed downstream KER degradation and operational incidents. The goal is to find a set of non-negative hazard weights (w) that best predict negative outcomes. Concurrently, the scalar `I` is mapped to the normalized `rcalib` coordinate using a piecewise-affine function that maps the corridor bands ($I_{\text{safe}}, I_{\text{gold}}, I_{\text{hard}}$) to the target risk ranges [0, 0.1], [0.1, 0.3], and [0.3, 1.0], respectively [7]. The corridor bands themselves are chosen as quantiles of the empirical distribution of `I` across windows labeled as "clean," "moderate," and "chronic" by domain experts [8]. This ensures that the numerical value of `rcalib` has a direct, interpretable connection to the quality of the ingested data.

For `rsigma`, the calibration follows a similar empirical path. The component risks—drift, noise, bias, and loss—are each defined with their own corridor bands derived from sensor calibration histories and maintenance logs [10]. The composite norm, $r_\sigma = \min\left(1, \sqrt{\sum_k \kappa_k^2 r_k^2}\right)$, combines these components into a single `rsigma` value [9]. The calibration effort involves regressing `rsigma` against known sensor failures and periods of poor data quality to determine the appropriate weighting vector $\boldsymbol{\kappa}$, where components like drift and bias are expected to carry more weight than noise and loss, reflecting their relative hazard [5]. The final step is to establish the global `rsigma` corridor bands (deep-safe ~0–0.1, gold ~0.1–0.3, hard $\geqslant$0.3) by aligning the distribution of computed `rsigma` values with observed system behavior, ensuring that the hard band acts as a reliable threshold for blocking deployments [3]. This entire calibration process is itself a research object, documented in `IngestErrorToRcalibMap2026v1.aln` and `RsigmaComponentCalibration2026v1.aln`, and its results are hex-stamped into the `DataQualityPlane2026v1.aln` specification, creating a fully traceable and auditable contract between the mathematical model and the empirical data it represents [6]. The resulting system, with its grammar-hardened plane, calibrated parameters, and executable invariants, raises the knowledge factor (K) towards 0.95 and keeps the risk-of-harm (R) in the low 0.12 band, with all remaining risk explicitly surfaced through the metrics themselves rather than hidden in implementation details [3].

Field / Invariant	Type / Description	Rationale
`plane_id`	`Utf8Id32`	Unique identifier for this data quality plane configuration (e.g., DQ-PHX-2026-01). [4]
`rsigma`	`RiskCoord01`	Normalized sensor uncertainty risk coordinate (0.0 to 1.0). [7]
`rcalib`	`RiskCoord01`	Normalized ingest/schema quality risk coordinate (0.0 to 1.0). [8]
`lyap_channel`	`EnumLyapChannel`	Must be 'dataquality', binding the plane to a dedicated Lyapunov channel. [10]
`corridor_table_id`	`Utf8Id64`	Foreign key to the master corridor table (e.g., ecosafety.corridors.v2). [9]
`rsigma_corridor_row`	`Utf8Id32`	VarID of the corridor row for rsigma bands within the corridor table. [3]
`rcalib_corridor_row`	`Utf8Id32`	VarID of the corridor row for rcalib bands within the corridor table. [5]
`lyap_weight_rsigma`	`Float64`	Non-negative weight for rsigma in the Lyapunov residual, Vt. [6]
`lyap_weight_rcalib`	`Float64`	Non-negative weight for rcalib in the Lyapunov residual, Vt. [3]
`lane`	`EnumLane`	Governing lane (RESEARCH, PILOT, PROD) for this plane's rules. [8]
`admissible_for_blockchain`	`Bool`	True if this plane config passes all proofs and monotonicity checks for on-chain commitment. [7]
`invariant.dq1.range`	Constraint	Enforces `0.0 <= rsigma, rcalib <= 1.0`. Prevents invalid risk values. [10]
`invariant.dq5.weights_nonneg`	Constraint	Enforces `lyap_weight_rsigma, lyap_weight_rcalib >= 0.0`. Guarantees data quality can only down-credit K/E/R. [9]
`invariant.dq7.proof_req`	Constraint	Requires existence of a `ker_proofs.v1` entry for monotonicity if `admissible_for_blockchain` is true. Ties on-chain trust to formal proof. [3]

This structured, invariant-driven approach to defining the data quality plane transforms it from a passive data structure into an active, intelligent contract that governs the safety and reliability of the entire ecosafety system.

Canonical Schemas and Compile-Time Enforcement via `derive(AlnShard)`

While the `DataQualityPlane2026v1.aln` defines the semantics of data quality, the `EcoNetSchemaShard2026v2.aln` establishes the bedrock of data representation: the canonical schema for all ecosafety qpudatashards. This ALN particle acts as the definitive source of truth for the mandatory and optional fields that constitute a valid, production-grade shard. It is the grammar that dictates what information must be present for a shard

to be considered trustworthy and eligible for promotion to higher lanes or on-chain storage. Complementing this static schema is the `derive(AlnShard)` procedural macro, a powerful tool that translates the declarative rules of ALN into imperative, compile-time enforcement within the Rust codebase. Together, these two elements create a system where invalid data structures are not merely rejected at runtime but are made unrepresentable in the language of the compiler, drastically reducing the surface area for bugs related to schema mismatches, missing fields, and incorrect data types. This section details the finalized structure of the `EcoNetSchemaShard2026v2.aln`, the responsibilities of the `derive(AlnShard)` macro, the strategy for handling schema evolution through new-write-only migrations, and the vision for extending this pattern into a type-safe governance lattice.

The `EcoNetSchemaShard2026v2.aln` is designed as the schema-of-schemas, consolidating the essential fields required across all ecosafety shard families, including Phoenix, Cyboquatic, and FOG. Its purpose is to enforce a consistent, comprehensive structure that captures not only the raw data but also the complete context of its evaluation, from ingestion to actuation. The columns in this ALN particle will include a mix of mandatory data-quality fields, KER metrics, and cryptographic provenance identifiers. Key mandatory fields will be `rcalib` and `rsigma` to ensure data quality is always tracked, followed by the full suite of trust scalars: `Dsensor`, `Ddata`, and `Dcombined` [2]. The KER window metrics must also be standardized, including `Kraw`, `Eraw`, `Rraw`, `Kadj`, `Eadj`, and `Rglobal`. Crucially, governance-related fields such as `lane` and `kerdeployable` are made mandatory, ensuring that every shard carries its compliance status. Finally, the schema must enforce the presence of cryptographic anchors, including `evidencehex`, `signinghex`, `ALNSPECHASH`, and `ALNSHARDID`, making them integral parts of the shard's identity and verifiability [4]. The development of this schema involves a census of existing live families to map their fields to this canonical header, resulting in a `ShardFamilyToCanonicalMatrix2026v1.csv` that documents the current state and migration requirements for each family [5].

The `derive(AlnShard)` macro is the engine that makes the `EcoNetSchemaShard2026v2.aln` actively secure. When a developer writes a Rust struct to represent a shard family (e.g., `PhoenixNodeShard`), they will derive the `AlnShard` trait from it. At compile time, the macro will parse the relevant rows from `EcoNetSchemaShard2026v2.aln` and generate a significant amount of boilerplate code tailored to that specific shard type. The primary responsibility of the macro is to implement the `validate_aln_contract` method. This method performs a multi-faceted check: first, it verifies that the Rust struct's field count and names perfectly match the mandatory columns defined in the ALN schema, failing the build if there is any

discrepancy ("no schema, no build") [6]. Second, it auto-implements the logic for generating `RiskVector` and computing the Lyapunov residual, `V_t`, ensuring consistency with the formula $V_t = \sum_j w_j r_j^2$ [3]. Third, it wires in the corridor-aware normalization for all `isriskcoord=true` columns, fetching the correct bands and normalization function from `EcoCoreParameters` and `EcoNetSchemaShard` to generate `RiskCoord` values in [0,1] [7]. Fourth, it enforces the cryptographic provenance fields, ensuring `evidencehex` and `signinghex` have the correct types and formats. Finally, it implements the "tightest band wins" logic, allowing a shard to reference multiple versions of the ALN contract and compiling only if the effective corridor bands for any overlapping coordinate are the strictest possible [8].

A critical aspect of managing a canonical schema over time is handling evolution without compromising the integrity of historical data. The proposed strategy is a "new-write-only" migration. Instead of attempting to edit or patch existing v1 shards, which would alter the immutable record stored on the blockchain, a new schema version (`v2`) is introduced. Old shards remain untouched, while new data is written to new shards conforming to the `v2` schema. This process involves creating back-filled or inferred values for the new mandatory fields in the `v2` schema. For example, if a `rcalib` field was previously missing, a migration script could calculate a baseline value based on the shard's lane or other available metadata, but this calculated `rcalib` would naturally be higher (worse) than a value derived from actual ingest diagnostics, thus correctly down-weighting the shard's trustworthiness. This approach is a monotone safety improvement: it either leaves KER gates unchanged or makes them stricter, but it never allows a lower-quality view to pass a gate that a higher-quality view would fail. The entire migration process is itself a research object, documented in `ShardMigrationRecipes2026v1.aln` and validated by replay scripts that rescore old and new versions of the data to confirm that production gates are not weakened [10].

Looking beyond simple validation, the principles behind `derive(AlnShard)` can be extended to create a powerful type-level safety lattice for governance. This involves generating a hierarchy of structs that encode ALN contracts and lanes as part of their type signature. For instance, the macro could generate a `NodePlacementRaw` struct from CSV input. Then, a `validate_aln_contract` call would produce a `NodePlacementValidated<AlnContractTag, Lane>` struct, which is only constructible if all ALN, corridor, and KER checks pass. Finally, the main deployment logic would accept as input only the most restrictive type, such as `NodePlacementValidated<Prod>`. This creates a static guarantee at compile time that no data can bypass the validation rules for its designated lane. A piece of code operating on a `PROD`-qualified shard cannot accidentally be passed a `PILOT`-qualified

shard, as the Rust type checker would reject the operation. This defense-in-depth strategy, combining ALN-driven normalization, Rust-level trait objects, and type-level tags, moves governance from a series of runtime checks into the very fabric of the program's type system, raising the knowledge factor (K) to approximately 0.96 and the eco-impact (E) to 0.92 by strongly reducing the potential for runtime configuration errors [9]. The residual risk-of-harm (R) is lowered to around 0.11, concentrated in the correctness of the macro implementation and the calibration of corridor bands, both of which are now surfaced and bounded by CI tests and shard-level KER scores [5].

Component	Responsibility	Impact on Safety and Reliability
`EcoNetSchemaShard2026v2.aln`	Defines the canonical, mandatory fields for all ecosafety shards, including `rcalib`, `rsigma`, `lane`, `kerdeployable`, and cryptographic anchors.	Creates a single source of truth for data structure, ensuring all critical safety and governance fields are present and standardized.
`derive(AlnShard)` Macro	Generates Rust code for schema validation, corridor-aware normalization, KER metric computation, and ALN contract enforcement at compile time.	Makes invalid states unrepresentable. Catches schema mismatches, missing fields, and incorrect types before runtime, drastically reducing bug surface.
New-Write-Only Migration	Introduces new schema versions (e.g., v2) alongside old ones (v1) without modifying historical data. Backfills new fields with conservative defaults.	Preserves the immutability and auditability of the blockchain ledger while incrementally improving data quality and coverage over time.
Type-Safe Governance Lattice	Extends the macro to generate types like `NodePlacementValidated<AlnContractTag, Lane>`, encoding ALN and lane info in the type system.	Provides static, compile-time guarantees that governance rules (e.g., lane-specific gates) are enforced, preventing accidental violations.

By combining a rigorously defined canonical schema with a powerful compile-time enforcement macro, this approach creates a deeply embedded and resilient layer of security that underpins every subsequent computational step in the ecosafety stack.

Cryptographic Provenance and Blockchain Governance Rules

The foundation of a truly hardened system rests not only on well-defined grammar and schemas but also on an unassailable chain of custody for every artifact involved. The

third pillar of this strategy is the establishment of a cryptographic provenance kernel, anchored by Decentralized Identifiers (DIDs), `evidencehex`, and `ALNSPECHASH`. This kernel ensures that every line of ALN code, every shard schema, every corridor definition, and every final qpudatashard is cryptographically tied to its creator and its precise version at creation time, making tampering detectable and authorship undeniable [3]. This cryptographic bedrock enables the fourth pillar: the formulation of clear, ALN-governed rules for what can be written onto the ALN-Blockchain. By moving beyond simply storing data and instead enforcing a set of on-chain admission policies, the ledger becomes an active participant in the safety and governance of the ecosystem. This section details the design of the `ProvenanceKernel2026v1.aln` and `CrateProvenance2026v1.aln` to manage cryptographic anchors, describes the `LedgerLaneRules2026v1.aln` to govern on-chain writes, and explains how these rules are enforced through CI gates and ledger-side validators, effectively turning the blockchain into a rule-enforcing device.

The cryptographic provenance kernel is built upon three core components. First, Bostrom DIDs serve as the human-readable, decentralized identities for developers, organizations, and automated systems responsible for creating artifacts. Every crate, ALN file, and shard is associated with a DID, establishing a clear chain of accountability [5]. Second, the `evidencehex` field is a 64-character hexadecimal digest that is deterministically computed from a canonical serialization of a shard's content. As defined by the `ProvenanceKernel2026v1.aln`, this serialization includes all mutable fields in a specific order (e.g., RFC4180 format), excluding the evidence and signature fields themselves. This means that any change to a non-excluded field will necessarily result in a different `evidencehex`, providing a powerful, immediate signal of modification [4]. Third, the `ALNSPECHASH` is a similar digest that anchors a shard to the specific ALN grammar (`EcoNetSchemaShard2026v2.aln`) and corridor bundle (`ecosafety.corridors.v2`) used to validate it. This ensures that a shard created under one set of rules cannot be retroactively validated under another. Together, these three elements—a DID for authorship, an `evidencehex` for content integrity, and an `ALNSPECHASH` for rule integrity—form a robust cryptographic seal around every artifact in the system.

To manage this web of dependencies and provenance claims, two new ALN particles are required. The first is `ProvenanceKernel2026v1.aln`, which formalizes the rules for generating `evidencehex` and `ALNSPECHASH`. This ALN will explicitly list the columns to be included or excluded from the `evidencehex` computation, specify the canonical byte encoding (e.g., UTF-8 with CRLF line endings), and define which ALN files contribute to the `ALNSPECHASH` [3]. This document becomes the authoritative guide for

all implementations, ensuring that Rust, C, and any other language producing shards computes the same digests. The second particle, `CrateProvenance2026v1.aln`, creates a registry mapping each core software crate to its Bostrom DID and its unique crate hex-stamp. This index links the logical identity of a piece of code to its physical implementation, allowing a validator to recompute the `evidencehex` of a shard and trace it back to the specific version of the software that produced it. This level of detail is crucial for debugging, auditing, and establishing long-term trust in the system's outputs [6]. CI systems can then run coverage jobs that intentionally flip individual bits in synthetic shards to verify that the `evidencehex` changes exactly when it should (for mutable fields) and remains stable when only immutable fields like `signinghex` are altered, providing a rigorous test of the provenance logic itself [7].

With the provenance of artifacts secured, the final step is to apply these principles to the governance of the ALN-Blockchain itself. Not all data or configurations should be allowed to be written to the immutable ledger. This requires a set of on-chain admission rules. The `LedgerLaneRules2026v1.aln` is designed to capture these rules. This ALN particle would define, for each lane (RESEARCH, PILOT, PROD) and potentially for each shard family, a precise set of conditions that must be met for a shard to be committed to the blockchain. For example, a rule for the PROD lane might state that a `PhoenixIngestTrust` shard is only admissible if its `kerdeployable` flag is true, its `rcalib` is below the gold band, it contains no deprecated fields, and its `evidencehex` corresponds to a known-good version of the `Phoenix ingest modules` crate listed in `CrateProvenance2026v1.aln`. These rules are not arbitrary; they are extensions of the safety contracts already defined in the `EcoNetSchemaShard` and corridor grammars. A shard that fails to meet the PROD lane rules for data quality or governance cannot be promoted, regardless of how clean its physical measurements appear [8].

These ALN-defined rules are enforced at multiple points in the workflow. First, the CI/CD pipeline must incorporate checks that validate incoming artifacts against the `LedgerLaneRules2026v1.aln`. Before a commit is accepted, a CI job will parse the shard and the rules, rejecting any combination that violates the policy. This acts as a pre-flight check, preventing non-compliant data from ever reaching the staging environment. Second, the ALN-Blockchain itself must have a smart contract or equivalent validation layer that re-evaluates these rules upon receipt of a write transaction. This ensures that even if a malicious actor were to compromise the CI system, the final arbiter of truth on the chain would still enforce the safety and governance policies. This dual-layer enforcement—pre-commit CI checks and post-commit on-chain validation—creates a robust defense-in-depth. It guarantees that the blockchain ledger contains only those artifacts that satisfy the most stringent requirements for trustworthiness, creating a permanent and immutable record of all production-grade, safety-certified data and

decisions. This systematic application of cryptographic anchoring and rule-based governance raises the knowledge factor (K) to approximately 0.95 and the eco-impact (E) to 0.91, as it prevents bad actors and corrupted data from polluting the trusted record, while keeping the risk-of-harm (R) in the 0.12 band, with the residual risk now residing in the calibration of the rules themselves, which is explicitly surfaced and subject to governance review [10].

Empirical Calibration and Replay-Based Verification Harnesses

A hard grammar and a canonical schema are necessary but insufficient foundations for a safe and reliable ecosafety system. The abstract mathematical constructs for `rcalib` and `rsigma` must be grounded in the messy reality of real-world data. The fifth pillar of this framework is a rigorous, empirical approach to calibration, coupled with automated, replay-based verification harnesses that turn mathematical proofs into executable, repeatable tests. This methodology ensures that the theoretical models of data quality accurately reflect their impact on system performance and that the system behaves as intended under a wide variety of fault conditions. The process begins with using Phoenix data to fit the weights and corridor bands of the ingest and sensor kernels. Once calibrated, a suite of specialized CI jobs, or "replay runners," is developed. These runners simulate faults—such as injecting synthetic ingest errors or sensor degradations—into historical datasets and automatically verify that the system's response aligns with the expected monotonic and Lyapunov-stability properties. This section details the empirical calibration studies for Phoenix, the design of the replay-based CI harnesses, and the numerical considerations for implementing the residual kernel, forming a closed loop of theory, calibration, and validation.

The calibration of the `rcalib` and `rsigma` kernels is a critical empirical task that bridges the gap between abstract diagnostics and tangible system performance. For `rcalib`, the process involves leveraging the rich history of Phoenix ingest logs. The first step is to instrument the ingest pipeline to generate detailed diagnostic vectors, counting the number of missing columns (`N_missing`), RFC4180 formatting errors (`N_rfc`), type mismatches (`N_type`), and varid conflicts (`N_varid`). These counts, along with their corresponding hazard weights (`w_miss`, `w_rfc`, `w_type`, `w_varid`), are used to compute the scalar ingest error `I` for each window [1]. The calibration study then proceeds by performing a regression analysis, correlating the computed `I` values with downstream consequences such as KER score degradation, incident reports, and manual

audits of data quality. The objective is to find a set of non-negative weights that maximizes the predictive power of `I` for negative outcomes. Simultaneously, the corridor bands for `I`—`(I_safe, I_gold, I_hard)`—are determined by finding the quantiles of the empirical distribution of `I` across windows that have been manually classified as "clean," "moderate," and "chronic." This ensures that the normalized `rcalib` coordinate, derived from `I` via a piecewise-affine map, has a clear, interpretable meaning: a `rcalib` in [0, 0.1] corresponds to clean data, [0.1, 0.3] to moderate issues, and >0.3 to chronic problems requiring intervention [7].

A similar empirical approach is used for `rsigma`. The component risks for drift, noise, bias, and loss are first calibrated by analyzing sensor calibration histories, lab references, and maintenance logs to establish their own `(safe, gold, hard)` bands [8]. The composite `rsigma` norm, $r_\sigma = \min\left(1, \sqrt{\sum_k \kappa_k^2 r_k^2}\right)$, is then tuned by regressing the full `rsigma` value against downstream failures or adverse ecological events. This helps determine the weighting vector $\boldsymbol{\kappa}$, ensuring that more hazardous components like bias and drift are given higher priority in the final risk calculation [9]. The final step is to calibrate the global `rsigma` corridor bands by aligning the distribution of `rsigma` values computed from historical data with observed system behavior. For instance, windows with `rsigma` consistently above 0.3 should correspond to periods of blocked actuation or flagged anomalies, validating that the hard band serves as an effective safety threshold [10]. All these calibration efforts, including the fitted weights, band edges, and regression models, are captured in research artifacts like `IngestErrorToRcalibMap2026v1.aln` and `RsigmaComponentCalibration2026v1.aln`, providing a transparent and auditable record of the calibration process.

With the kernels calibrated, the next step is to build a suite of replay-based CI harnesses to provide strong guarantees about the system's behavior. These are not traditional unit tests but rather end-to-end behavioral property tests. One key harness is the **Phoenix Ingest Failure Atlas**. This runner takes a historical ingest window, synthesizes various fault scenarios (e.g., artificially introducing a missing column, corrupting a type, or adding a rogue varid), and reprocesses the modified data. The CI job then asserts a set of expected inequalities: the newly computed `rcalib`, Lyapunov residual `V_t`, and max-risk `R` must be weakly increasing or flat, while the adjusted eco-impact `Eadj` and knowledge `Kadj` must be weakly decreasing or flat [3]. Any violation of these monotonicity properties constitutes a regression and blocks the merge, automatically catching bugs in the normalization or KER logic. An analogous **Rsina/Rcalib Forensics**

Runner would inject synthetic increases in sensor drift, noise, or bias to verify the same properties for the `rsigma` coordinate.

Another critical harness is the **Spec-Upgrade Replayer**. When a corridor is tightened—for example, by moving a `hardmax` band inward—the system's behavior must become more conservative. This runner verifies the "tightest-band-wins" property by recomputing the risk coordinates for a large sample of historical raw data under both the old and new corridor definitions. The CI check asserts that for every data point, the new risk coordinate is greater than or equal to the old one. This proves that tightening governance rules can only raise risk scores, lower K/E, and increase `V_t` and `R`, ensuring that policy changes are inherently conservative and never make a faulty view appear safer than it was before [6]. Finally, the **Lyapunov Proof Harness** runs long replay simulations of the system, checking that the discrete Lyapunov condition, $V_{t+1} \leq V_t$, holds outside a small interior band, as required by the safestep invariant. These replay harnesses, which produce their own evidence-bearing qpudatashards upon execution, transform the mathematical invariants into a living, automated test suite that continuously validates the system's safety properties against empirical data.

Finally, the computational kernel that executes these calculations must be numerically robust. The plan is to implement dual-profile fixed-order residual kernels. An "embedded profile" uses stack-allocated arrays optimized for low-latency, real-time decision-making, while an "atlas profile" is a batch-oriented implementation for massive-scale offline audits. Both profiles must agree on the result. The `safestep_satisfied` function, which evaluates the Lyapunov condition, must account for floating-point imprecision. It will use a set of very small tolerances: a relative tolerance (`vt_rel_tol`) of ~1e-6 and an absolute tolerance (`vt_abs_tol`) of ~1e-9 [5]. The function will accept a step if `next.vt` is within this tight tolerance band of `prev.vt`. Deriving worst-case floating-point error bounds for the typical number of risk coordinates (e.g., N=16) would further justify these tolerance choices, providing a formal guarantee of the kernel's numerical stability and reliability.

Verification Harness	Purpose	Input	Behavioral Property Asserted
Phoenix Ingest Failure Atlas	Validate that increased ingest faults lead to predictable safety degradation.	Historical Phoenix ingest shards, synthetic fault injections (missing columns, etc.).	`rcalib`, `V_t`, `R` weakly increase; `Kadj`, `Eadj` weakly decrease.
Rsina/Rcalib Forensics Runner	Validate that increased sensor uncertainty leads to predictable safety degradation.	Historical Phoenix sensor shards, synthetic uncertainty injections (drift, noise, etc.).	`rsigma`, `V_t`, `R` weakly increase; `Kadj`, `Eadj` weakly decrease.
Spec-Upgrade Replayer	Verify that tightening corridor bands is a conservative policy change.	Historical raw data, pairs of old and new corridor definitions.	For all data points, `risk_new >= risk_old`. Ensures "tightest band wins".
Lyapunov Proof Harness	Check that the discrete Lyapunov condition holds over long simulations.	Long time-series of historical shard data.	`V_{t+1} <= V_t` outside a small interior band, within numerical tolerances.

This combination of empirical grounding and rigorous, automated verification forms a powerful feedback loop that ensures the data quality plane is not only theoretically sound but also practically effective at lowering the system's risk-of-harm.

Mathematical Foundations: Lyapunov Stability and Monotonicity Proofs

The safety guarantees of the ecosafety stack rest upon a solid mathematical foundation. While empirical calibration and automated testing provide strong evidence of correct behavior, they do not replace the need for formal proofs of the system's core properties. This section details the essential mathematical lemmas that must be formally established and linked to the ALN grammar to create an executable contract for safety. These proofs cover the non-negativity and convexity of the extended Lyapunov residual, V_t^{full}, which now includes terms for `rcalib` and `rsigma`. They also cover the monotonicity of the risk coordinates and the KER metrics with respect to their underlying diagnostics, ensuring that data quality can only degrade performance, never improve it. Finally, they address the "tightest-band-wins" property of corridor tightening, proving that policy changes are conservative. Each of these lemmas is treated as a distinct mathematical object, serialized as an ALN row in `ecosafety.ker_proofs.v1`, and linked to the relevant `DataQualityPlane2026v1` specification. This transforms abstract mathematical requirements into concrete, verifiable, and cryptographically anchored commitments that are checked at compile time and on-chain.

The first set of proofs pertains to the extended Lyapunov residual, defined as:

$$V_t^{full} = \sum_{j \in \mathcal{P}} w_j r_{j,t}^2 + w_{\text{calib}} r_{\text{calib},t}^2 + w_\sigma r_{\sigma,t}^2$$

where $\mathcal{P}$ is the set of all physical risk coordinates, and $w_j, w_{\text{calib}}, w_\sigma \geq 0$ are the corresponding non-negative Lyapunov weights [10]. The first lemma, **Lemma A: Non-negativity and Convexity**, establishes that V_t^{full} is a valid Lyapunov-like function. Since it is a sum of squared terms multiplied by non-negative weights, it is manifestly non-negative: $V_t^{full} \geq 0$ for all t. Furthermore, $V_t^{full} = 0$ if and only if all risk coordinates, including `rcalib` and `rsigma`, are simultaneously zero. This proves that the origin (perfect data quality and perfect physical state) is the unique minimum of the residual. As a function of the risk vector \mathbf{r}<em _text_int="\text{int">t, V_t^{full} is a positive semidefinite quadratic form, which implies it is convex. This convexity is a crucial property for stability analysis [7]. The second related property is the **Discrete Lyapunov Condition**, which is the heart of the `safestep` contract. It states that there exists an interior sublevel set $V</em>$, then applying a }} $suchthatiftheresidualexceedsthisthreshold, i.e.,$V_t^{full} > V_{\text{int}}$ `safestep` control action must result in a non-expansive residual: $V_{t+1}^{full} \leq V_t^{full}$ [8]. This proves that the system is dissipative and tending towards a safe region of the state space whenever it ventures too far out. Including `rcalib` and `rsigma` in this residual strengthens the proof, as data quality issues are now a direct contributor to the overall system energy.

The second major set of proofs concerns monotonicity. These lemmas are crucial because they formalize the intuitive idea that "more errors can only hurt." The first monotonicity lemma, **Lemma B: Monotonicity vs Diagnostics**, states that any increase in the underlying diagnostics or uncertainties can only cause the risk coordinates and the Lyapunov residual to increase or stay the same. Specifically, the ingest scalar I is a linear function with non-negative weights, so it is monotone non-decreasing in each diagnostic count $(N_{\text{miss}}, N_{\text{rfc}}, \ldots)$. The piecewise-affine corridor map is also monotone non-decreasing, which implies that `rcalib` is a monotone non-decreasing function of I, and therefore of each individual diagnostic count. Similarly, the `rsigma` composite norm is a monotone non-decreasing function of each of its component risks $(r_{\text{drift}}, r_{\text{noise}}, \ldots)$. Since `rcalib` and `rsigma` are monotone non-decreasing, and they enter the quadratic terms in V_t^{full} with non-negative weights, it follows that V_t^{full} is a monotone non-decreasing function of all underlying diagnostics and sensor uncertainties [9]. This

property is what allows the replay-based CI harnesses to assert that injecting faults must lead to a weakly larger residual.

Building on this, the third set of lemmas formalizes the behavior of the Knowledge, Eco-Impact, and Risk (KER) metrics. The **Trust Scaling** formulas are $D_{\text{data}}=1-r_{\text{calib}}$ and $D_{\text{combined}}=D_{\text{sensor}}\cdot D_{\text{data}}$, leading to the adjusted KER scores $K_{\text{adj}}=K_{\text{raw}}\cdot D_{\text{combined}}$ and $E_{\text{adj}}=E_{\text{raw}}\cdot D_{\text{combined}}$ [10]. The **Lemma D: Kadj, Eadj non-increasing in rcalib** provides the formal justification for the "only-downcredit" rule. Taking the partial derivative with respect to `rcalib`, we get $\frac{\partial K_{\text{adj}}}{\partial r_{\text{calib}}}=K_{\text{raw}}\cdot D_{\text{sensor}}\cdot(-1)\leq 0$, since all terms in the product are non-negative. This proves that for a fixed raw K/E and sensor trust, worse ingest quality (`higher rcalib`) can only decrease or leave `Kadj` and `Eadj` unchanged. The maximum risk `R` is defined as $R=\max_{j,t} r_{j,t}$ over all coordinates, including `rcalib` and `rsigma`. The **Lemma E: R and E monotone vs rcalib/rsigma** follows directly: any increase in `rcalib` or `rsigma` at any point in time can only increase or leave `R` unchanged, and therefore can only decrease or leave `E = 1-R` unchanged [7]. These monotonicity properties are currently described in prose and must be pinned as machine-checkable ALN invariants (e.g., `onlydowncreditctx,t`) and hex-stamped lemmas.

Finally, the **Corridor Tightening Lemmas** formalize the conservative nature of governance updates. Let $r(x|\theta)$ denote the corridor-normalized risk of a raw scalar x under band parameters $\theta=(x_{\text{safe}},x_{\text{gold}},x_{\text{hard}})$. The **Lemma F: Tightest-band-wins** states that if θ' is a "tightening" of θ (meaning $x'_{\text{safe}}\leq x_{\text{safe}}$, $x'_{\text{gold}}\leq x_{\text{gold}}$, and $x'_{\text{hard}}\leq x_{\text{hard}}$, with at least one strict inequality), then for any fixed raw value x, the resulting risk is weakly larger: $r(x|\theta')\geq r(x|\theta)$ [8]. This is a direct consequence of the piecewise-affine mapping. If x was in the safe band under θ, it is almost certainly in the safe band under θ', so r remains 0. If x was in the tail of the gold band under θ, moving the gold and hard bands inward is likely to push it into the hard band, causing r to jump to a value closer to 1. This property must be enforced in spec-upgrade CI, ensuring that tightening bands never yields a lower risk score for the same physical error [5]. Each of these lemmas—convexity, monotonicity, and corridor tightening—is a small, discrete mathematical object that can be represented in the ALN, providing an unambiguous, formal basis for the system's safety claims.

Lemma	Statement	Mathematical Formulation	Significance for Safety
Non-negativity & Convexity	The full Lyapunov residual V_t^{full} is always non-negative and convex in the risk vector.	$V_t^{full} \geq 0$ and is a positive semidefinite quadratic form.	Establishes V_t^{full} as a valid energy function for stability analysis, with a unique minimum at the safe state.
Monotonicity vs. Diagnostics	Increasing any underlying diagnostic count or sensor uncertainty can only increase or leave unchanged `rcalib`, `rsigma`, `V_t`, and `R`.	If $\mathbf{d}' \geq \mathbf{d}$, then $V_t^{full}(\mathbf{d}') \geq V_t^{full}(\mathbf{d})$.	Guarantees that data quality can only down-credit the system. This is the core property verified by ingest/sensor replay jobs.
KER Monotonicity	Adjusted KER scores `Kadj` and `Eadj` are non-increasing in `rcalib`; `R` is non-decreasing in `rcalib` and `rsigma`.	$\frac{\partial K_{\text{adj}}}{\partial r_{\text{calib}}} \leq 0$ and $\frac{\partial R}{\partial r_{\text{calib}}} \geq 0$.	Formalizes the "only-downcredit" rule. Proves that trust scaling works as intended, preventing over-claiming of K/E.
Tightest-Band-Wins	Tightening corridor bands (moving them inward) can only yield a higher or equal risk score for a fixed raw input.	If $\theta' \leq \theta$, then $r(x\|\theta') \geq r(x\|\theta)$.	Ensures that governance updates are conservative. Prevents a flawed view from appearing safer after a policy change.

By formalizing these properties and integrating them into the ALN as provable, hex-stamped facts, the system gains a layer of assurance that goes beyond empirical testing alone, creating a truly mathematically-grounded safety framework.

Actionable Roadmap: From Phoenix to Expanded Governance

This research culminates in a concrete, actionable roadmap for implementing the grammar-first hardening framework. The strategy is phased, beginning with the immediate priorities that offer the highest leverage for lowering risk-of-harm in the existing Phoenix ecosystem, and then expanding horizontally to other ecological domains and governance capabilities. The initial phase focuses on freezing the core ALN artifacts (`DataQualityPlane2026v1.aln`, `EcoNetSchemaShard2026v2.aln`), building the `derive(AlnShard)` enforcement macro, and launching the empirical calibration and replay-based CI efforts for Phoenix data. The subsequent phases involve extending this successful model to Cyboquatic and FOG families, strengthening the blockchain governance layer, and planning for future capabilities like social-license planes. This entire roadmap is itself managed as a research project, with each major task documented as a `KerResearchActionShard2026v1.csv` that tracks the knowledge factor (K), eco-impact (E), and risk-of-harm (R) for the work, ensuring the entire initiative remains aligned with the project's overarching goals [3] [5].

Phase 1: Foundational Hardening for Phoenix (Immediate Priority)

The immediate focus must be on establishing a hardened, self-validating ecosafety spine for the Phoenix basin. This is the highest-leverage activity because it directly impacts the residual risk-of-harm (currently 0.12) and provides a template for all future expansion.

- **Task 1.1: Finalize and Freeze Core ALN Grammars.** The `DataQualityPlane2026v1.aln` and `EcoNetSchemaShard2026v2.aln` specifications must be completed and frozen. This involves incorporating all the detailed columns, mandatory flags, and, most importantly, the full set of ALN invariants that govern data quality, corridor usage, and lane rules. This artifact becomes the official, immutable contract for all Phoenix-related data.
- **Task 1.2: Implement the `derive(AlnShard)` Macro.** The procedural macro must be developed to parse the finalized ALN files and generate Rust code for schema validation, corridor-aware normalization, and KER metric computation. Integration tests should be created to deliberately introduce schema mismatches and verify that the macro correctly fails compilation, proving its effectiveness.
- **Task 1.3: Launch Phoenix Calibration Study.** A dedicated calibration study should be initiated to empirically fit the weights and corridor bands for `rcalib` and `rsigma` using historical Phoenix data. This involves regression analysis of ingest faults against KER degradation and correlation of sensor component risks with known failures. The results will be stored in new ALN particles like `IngestErrorToRcalibMap2026v1.aln`.
- **Task 1.4: Develop Replay-Based CI Harnesses.** The "Phoenix Ingest Failure Atlas" and "Rsina/Rcalib Forensics Runner" CI jobs must be built. These harnesses will take historical Phoenix data, synthetically introduce faults, and assert the expected monotonicity and Lyapunov properties. Passing these tests will become a mandatory prerequisite for merging any changes to the ingest, sensor, or KER logic.

Phase 2: Expansion and Schema Evolution (Medium-Term Priority)

Once the Phoenix foundation is solid, the framework can be expanded to other domains and improved over time.

- **Task 2.1: Extend Governance to Cyboquatic and FOG.** The proven grammar and enforcement pattern from Phoenix should be applied to Cyboquatic fouling/biopack and FOG/routing shard families. This involves creating ALN particles that import the Phoenix data quality plane, making `rcalib` and `rsigma` mandatory for these basins as well, and extending `kerdeployable` and placement logic to be gated by the unified data-quality plane.
- **Task 2.2: Execute New-Write-Only Migrations.** A formal process for migrating existing v1 Phoenix, Cyboquatic, and FOG shards to the new `v2` canonical schema must be designed and executed. This "new-write-only" strategy will create new v2

shards alongside old v1 ones, back-filling mandatory fields with conservative estimates and calculating a more accurate `rcalib` for older data without altering the original, immutable artifacts [6].

- **Task 2.3: Calibrate Domain-Specific Corridors.** While importing the general data quality plane, each domain should also develop its own specialized corridor bands for its unique risk coordinates (e.g., `rfouling`, `rbio` for Cyboquatic). These local corridors should be tied back to the global eco-impact scoring to maintain consistency across the ecosystem.

Phase 3: Strengthening the Blockchain Layer and Numerical Kernel (Ongoing Priority)

Parallel to schema and governance work, the underlying technical components must be hardened.

- **Task 3.1: Formalize Cryptographic Provenance.** The `ProvenanceKernel2026v1.aln` and `CrateProvenance2026v1.aln` must be finalized. CI jobs should be added to run coverage tests that prove the `evidencehex` logic correctly detects all permissible changes and ignores impermissible ones, solidifying the cryptographic foundation [3].
- **Task 3.2: Define On-Chain Admission Rules.** The `LedgerLaneRules2026v1.aln` should be developed to specify precisely which shard types and KER scores are admissible in the PROD lane. CI and on-chain validators must be updated to enforce these rules, turning the blockchain into an active safety gate.
- **Task 3.3: Implement Dual-Profile Fixed-Order Kernels.** The residual kernel must be implemented in two profiles: an "embedded" version for real-time `safestep` decisions and an "atlas" version for large-scale offline audits. CI property tests will ensure both profiles produce identical results within defined floating-point tolerances, guaranteeing numerical reproducibility [7].

Phase 4: Future Capabilities and Meta-Governance (Long-Term Vision)

The ultimate vision is a self-documenting, self-validating governance system that can evolve safely.

- **Task 4.1: Design Social-License and Equity Planes.** The success of the data quality plane provides a blueprint for adding new risk dimensions. A "social-license" plane, with a normalized coordinate `r_equity`, can be designed to measure equity and benefit-sharing metrics. This new plane would be integrated into the Lyapunov

residual and KER in the same way as `rcalib` and `rsigma`, creating a multi-axis risk assessment.

- **Task 4.2: Establish Cross-Language Read-Only Contracts.** To support diverse client ecosystems, a formal read-only contract should be established. This contract would mandate that all non-Rust clients can only consume qpudatashards and display KER residuals computed by the trusted Rust/ALN core; they are forbidden from re-computing kernels or stripping cryptographic signatures [5].
- **Task 4.3: Automate the Research Plan.** The entire roadmap should be managed within the system itself. Each task in Phases 1-4 will be represented as a `KerResearchActionShard2026v1.csv` row, capturing its K/E/R scores and linking to the relevant ALN specs, code, and CI jobs. This makes the project plan an auditable, on-chain artifact, enabling AI tools to reason about the system's evolution and ensuring that all growth is "just-by-researching-it" aligned [6].

By following this phased roadmap, the project can systematically build a deeply embedded, mathematically-grounded, and cryptographically-anchored safety framework. The initial focus on grammar hardening and empirical validation for Phoenix will pay dividends across the entire ecosafety stack, progressively lowering the risk-of-harm and increasing the trustworthiness of the system for decades to come. The final risk-of-harm, estimated at 0.12, is now tightly confined to the calibration of corridor bands and the correctness of the core mathematical proofs, risks that are explicitly surfaced and rigorously tested rather than hidden in complex runtime logic.

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